Student Dropout Prediction Using Machine Learning

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1. Introduction:

Student dropout is one of the major challenges faced by educational institutions worldwide. Students may discontinue their studies due to several factors such as poor academic performance, low attendance, stress, lack of motivation, financial problems, or personal issues. Early identification of students at risk of dropping out can help institutions provide timely support and improve overall student retention rates.

This project focuses on developing a machine learning-based prediction system that can identify whether a student is likely to drop out based on academic, behavioral, and demographic features.

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2. Objective:

The primary objective of this project is to build and compare multiple machine learning classification models to accurately predict student dropout risk.

The project also aims to:

- Perform detailed Exploratory Data Analysis (EDA)
- Handle missing and inconsistent data
- Apply feature engineering and feature selection techniques
- Improve model performance through hyperparameter tuning
- Save and test the trained model on unseen data.

3. Data Description:

Source:

Kaggle Dataset

Dataset Link:

https://www.kaggle.com/datasets/meharshanali/student-dropout-prediction-dataset

Dataset Information:

The dataset contains student-related academic, demographic, and behavioural information used to predict dropout status.

Features:

The dataset contains various student-related attributes such as:

- Gender
- Age
- Attendance Percentage
- GPA
- Study Hours
- Internet Access
- Family Income
- Stress Level
- Sleep Hours
- Participation in Activities
- Previous Academic Performance
- Dropout Status (Target Variable)

Target Variable:

- Dropout (Yes/No)

4. Data Collection:

The dataset was downloaded from Kaggle and imported into Google Colab using Pandas.

During data collection:

- Dataset structure was analyzed
- Data types were identified
- Missing values and duplicate records were checked
- Initial statistical analysis was performed

Libraries used:

- Pandas
- NumPy
- Matplotlib

- Seaborn
- Scikit-learn

5. Data Preprocessing - Data Cleaning:

Data preprocessing was one of the most important stages of the project.

Missing Value Handling

Missing values were identified using:

- isnull().sum()

Imputation techniques used:

- Median imputation for numerical columns
- Mode imputation for categorical columns

Duplicate Handling

Duplicate rows were detected using:

- duplicated().sum()

Duplicates were removed to improve data quality.

Outlier Detection

Outliers were detected using:

- Boxplots
- IQR (Interquartile Range) method

Extreme values were handled appropriately.

Skewness Handling

Skewed numerical distributions were analyzed and transformed when necessary.

6. Exploratory Data Analysis (EDA):

EDA was performed to understand patterns, relationships, and distributions in the dataset.

Visualizations Used:

- Histogram

- Boxplot
- Pair Plot
- Heatmap Correlation
- Pie Chart
- Bar Plot
- Count Plot
- Line Plot
- Kernel Density Estimation (KDE)
- Feature Importance Plot

Key Insights:

- Students with low GPA had higher dropout probability
- High stress levels negatively affected retention
- Students with poor attendance were more likely to drop out
- Academic performance strongly influenced dropout prediction
- Random Forest feature importance identified GPA and attendance as major features

7. Feature Engineering:

Categorical features were encoded using:

- Label Encoding
- One-Hot Encoding

Additional derived features were also created to improve model performance.

8. Feature Selection:

Feature selection techniques were applied to identify the most important variables affecting student dropout.

Methods Used:

- Correlation Analysis
- Random Forest Feature Importance
- SelectKBest

Irrelevant features were removed to improve model efficiency.

9. Split Data into Training and Testing Sets:

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

This helped evaluate the model's performance on unseen data.

10. Feature Scaling:

Numerical features were scaled using:

- StandardScaler

Feature scaling improved the performance of distance-based algorithms such as:

- KNN
- SVM
- Logistic Regression

11. Build the ML Model:

Multiple machine learning classification algorithms were implemented and compared.

Models Used:

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier
4. K-Nearest Neighbors (KNN)
5. Gradient Boosting Classifier
6. Support Vector Machine (SVM)

12. Model Evaluation:

The models were evaluated using classification metrics.

Evaluation Metrics:

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve

- AUC Score

Best Performing Model
Random Forest and Gradient Boosting achieved the highest accuracy among all models.

13. Hyperparameter Tuning and Pipeline:

Hyperparameter tuning was performed using:

- GridSearchCV

Parameters such as:

- Number of estimators
- Maximum depth
- Minimum samples split

were optimized to improve model accuracy.

Pipelines were also used to streamline preprocessing and training workflows.

14. Save the Model:

The best-performing model was saved using:

- Joblib

Saved File:

- dropout_model.pkl

15. Test with Unseen Data:

The trained model was tested using unseen student data to evaluate real-world prediction capability.

The model successfully predicted dropout risk with high accuracy.

16. Conclusion:

This project successfully developed a machine learning model for predicting student dropout risk.

Key findings include:

- Attendance and GPA were highly influential features
- Stress level and study habits significantly impacted dropout probability
- Ensemble models like Random Forest performed better than basic classifiers

- Proper preprocessing and feature selection significantly improved model performance

Limitations:

- Dataset size was limited
- Real-world educational data may contain more complex behavioral factors
- Class imbalance may affect prediction performance

17. Future Work:

Future improvements may include:

- Applying Deep Learning techniques
- Using real-time institutional data
- Deploying the model as a web application using Streamlit or Flask
- Handling class imbalance using SMOTE
- Adding more behavioural and psychological attributes
- Integrating real-time student monitoring systems